VICTORIA JUNIOR COLLEGE

JC 2 PRELIMINARY EXAMINATION 2017

NAME : ____

CT CLASS : ____

H2 BIOLOGY **9744/4**

Paper 4 **2 hours 30 minutes**

1. Write your name and CT group in the spaces at the top of this page.
2. Write in dark blue or black pen. You may use an HB pencil for any diagrams or graphs.
3. Answer all questions in the spaces provided on the Question Paper.
4. Students with the microscope and slide **must start with Question 2b first.**
5. The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.
6. At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [ ] at the end of each question.

For Examiner's Use	
1	
2	
3	
Total	

This document consists of **13** printed pages, excluding the cover page.

[Turn over

Answer **all** questions.

1. K1 is an extract prepared from the digestive tract of a small mammal. You are required to investigate the ability of this extract to bring about the breakdown of protein at varying pH, and determine the pH of an unknown solution through your observations.

Proceed as follows:

1. You have been supplied with a source of protein in the form of coagulated ('hard-boiled') egg which have been stained red.

2. Label five test-tubes **A, B, C, D** and **E** respectively. Prepare a beaker to act as a water-bath. The temperature of the water should be about 40˚C. It is not necessary to maintain this temperature.

To tube **A** add 2 cm^3 of K1 and 1 cm^3 of buffer, pH **2**.
To tube **B** add 2 cm^3 of K1 and 1 cm^3 of buffer, pH **4**.
To tube **C** add 2 cm^3 of K1 and 1 cm^3 of buffer, pH **6**.
To tube **D** add 2 cm^3 of K1 and 1 cm^3 of buffer, pH **8**.
To tube **E** add 2 cm^3 of K1 and 1 cm^3 of buffer of unknown pH, **X**.

3. Using a ruler and a scalpel, cut five pieces of egg of dimensions **2 cm in length, 0.5 cm wide and 0.5 cm in depth**. Place a piece of the egg in each of tubes **A, B, C, D** and **E**.

4. Place the five tubes in the water-bath at about 40˚C for 20 minutes while you continue with the rest of the question. Gently shake the tubes at five minute intervals during this time.

5. After 20 minutes, return to tubes **A, B, C, D** and **E**. Place a rubber bung in tube **A** and shake the tube vigorously ten times. Repeat this procedure on tubes **B, C, D** and **E.**

6. Allow the contents of the tubes to settle and then observe them carefully.

(a) **(i)** Record your observations in a suitable format in the space below, noting particularly any differences that you observe in the appearance of the contents of the five tubes. [6]

(ii) Based on your observation in (a) (i), estimate the pH of buffer **X**. [1]

(b) State the conclusions that you can draw, at this stage, about the action of K1 on the protein at different pH. Explain how your observations allow you to make these conclusions. [4]

(c) **(i)** Discuss two sources of error, other than temperature, that may have affected the accuracy of your results. [2]

(ii) Suggest improvements to reduce the sources of error that you have identified in **(c) (i).** [2]

(d) In a separate experiment, the effect of pH on potato catalase activity was studied using the following setup (shown in **Fig. 1**).

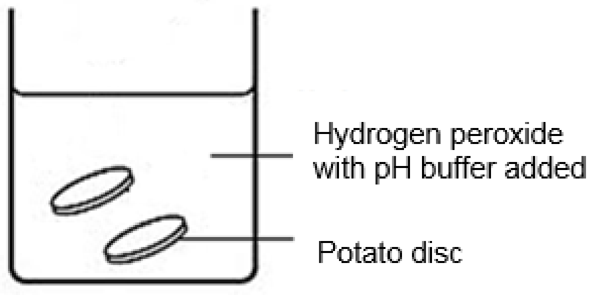

Fig. 1

Equal volume of hydrogen peroxide solution was added with buffer solution of varying pH into five beakers. Two potato discs were placed together into each beaker and the time taken for each disc to rise was recorded. Results obtained are shown in **Table 1**.

Table 1

pH buffer	**Time taken for potato disc to rise / s**		**Rate of reaction / s^{-1}**
	Disc 1	**Disc 2**	
4	58	63	
5	43	47	
6	41	42	
7	39	40	
8	44	49	
9	56	63	

(i) Calculate the rate of breakdown of hydrogen peroxide for each pH and record your answers in **Table 1**. (Working is not required) [1]

(ii) Plot a graph showing the effects of varying pH buffer on the rate of breakdown of hydrogen peroxide in the grid provided below. [4]

(iii) Explain how one can determine that the time taken for potato discs to rise at two different pH is significantly different from each other? [3]

[Total: 23]

2. **(a)** You are provided with an extract **P** from plant cells which contains a mixture of different carbohydrates.

You are required to identify which carbohydrates present in **P** can pass through the Visking tubing.

Proceed as follows:

1. Using the apparatus and reagents provided, carry out relevant tests to identify all the carbohydrates present in extract **P**.

(i) Use the space below to record the tests that you have performed, your observations and conclusions in a suitable format. [4]
Details of the tests are not required.

2. Prepare the Visking tubing by tying a knot at one end of the tubing.

3. Add 10 cm^3 of solution **P** into the Visking tubing and wash the outside of the tubing with water.

4. Place the Visking tubing into a boiling tube. Fold the open end over the top of the tube and secure it with a rubber band.

5. Add distilled water into the boiling tube. Leave the setup for 10 minutes.

6. After 10 minutes, remove the Visking tubing from the boiling tube and test the contents in the boiling tube for any carbohydrate molecules using the reagents provided.

(ii) What do your results indicate about the property of the Visking tubing? Explain your answer. [3]

(iii) Explain how you can modify the experiment to determine the rate of diffusion of the carbohydrate that you have identified in **(a) (ii)** above. [3]

(b) Slide **DB** shows a stained longitudinal section of a young root tip.

Examine **DB** carefully using low and high power objectives of your microscope. Note the occurrence and distribution of different kinds of cells in this section.

(i) Make a plan drawing of the entire section, within the outline drawn in **Fig. 2** below to show the different **regions**. These regions result from differences in the *shapes, sizes* and *structure* of the cells as well as in the frequency of mitosis.
Do not draw individual cells. Ignore the cells that make up the root cap region.

Annotate your drawing as fully as possible to describe the features of the cells in each region that you map. [5]

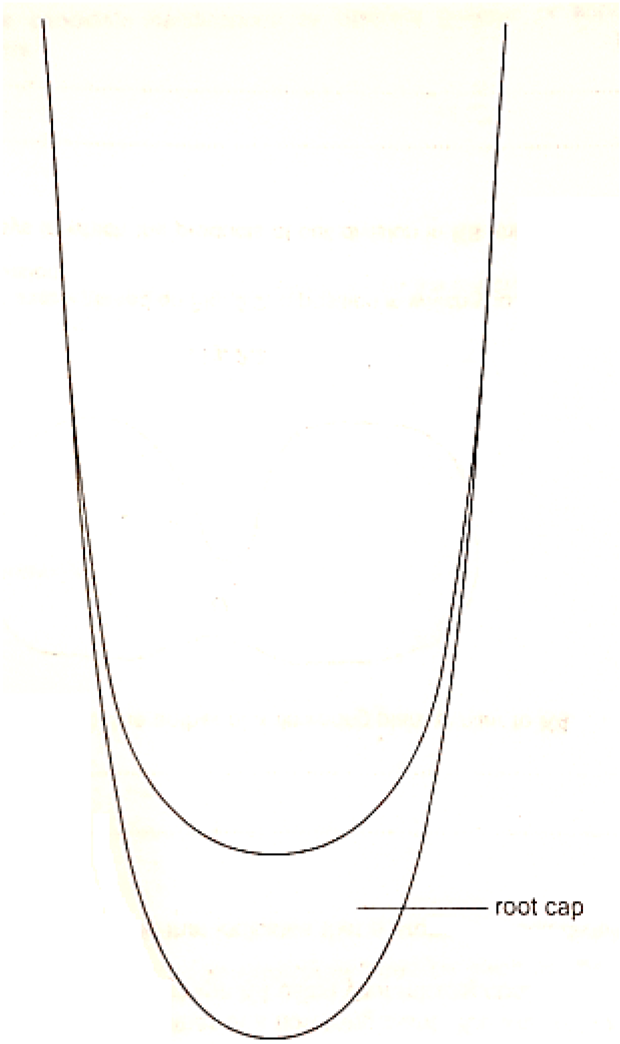

Fig. 2

(ii) In the space below, draw to the **same scale**, two cells that are at different stages of mitosis. Identify the stages and label the distinctive features of each stage in your drawings. [5]

[Total: 20]

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3. Yeast cells have transport proteins in their cell membranes for the uptake of nutrients from the surroundings. There are separate transport proteins for glucose and for maltose. When exposed to both glucose and maltose, the transport protein for maltose is down-regulated and is not produced.

Plan an investigation to find out whether or not the yeast transport proteins for glucose and maltose function at the same rate. The hypothesis is that the rate of uptake of glucose is higher than the rate of uptake of maltose.

You are provided with the following materials, which you can choose from. You may not use any additional materials and apparatus.

- an unlimited supply of 10% yeast suspension
- an unlimited volume of 10 g dm^{-3} glucose solution
- an unlimited volume of 10 g dm^{-3} maltose solution
- Benedict's solution
- dilute hydrochloric acid
- sodium hydrogen carbonate
- distilled water
- beakers and flasks of different sizes
- stop watch
- colorimeter and cuvettes
- centrifuge and centrifuge tubes
- thermometer
- thermostatically-controlled water baths
- pipettes and pipette fillers
- burettes and burette stands
- filter funnels and filter paper
- syringes
- Bunsen burner
- glass rods
- test-tubes and boiling tubes

Your plan should have a clear and helpful structure to include:
- an explanation of the theory to support your practical procedure
- a description of the method used including the scientific reasoning behind the method
- how you will record your results and ensure that they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- relevant risks and precautions taken
- The correct use of scientific and technical terms

[Total: 12]